

## Week 3 — Q2 in Depth: From Population to Analytic Cohort

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES · previews L2

## Week 3

Q2 in depth: from population to analytic cohort

**A cohort is the first result of an analysis.**

Today we follow one study from a Methods paragraph to a dataset that can be analysed.

## Q2, in depth — and this week's division of labour

Week 2 gave you six questions. **Today is Q2:** who is the population, and how is eligibility operationalised?

|                        | What it teaches                                                                                                 | On which paper          |
|------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Tuesday (today)</b> | the <b>judgment</b> —<br>turning a prose<br>eligibility paragraph into<br>logic you can defend                  | MASLD (Kueh)            |
| <b>Thursday (L2)</b>   | the <b>mechanics</b> —<br>multi-cycle download,<br>mortality linkage,<br>eligibility steps, analytic<br>dataset | benzodiazepine / opioid |
| <b>P1</b>              | both, transferred                                                                                               | your locked paper       |

- Framing question: *“If two teams read the same methods paragraph, would they build the same cohort?”*  
*Eligibility decisions are analysis choices. That is why they sit inside Q2, not in a housekeeping appendix.*

## What you should be able to do by the end

Given a published Methods section, you should be able to:

1. identify the files and variables needed to build the cohort
2. translate each inclusion rule into executable logic
3. classify every reduction in  $N$  by its cause
4. document ambiguities and show what alternative decisions would change

**The standard is a defensible audit trail, not an unexplained match to the paper.**

## The complete cohort-building sequence

| Step | Question                                           |
|------|----------------------------------------------------|
| 0    | Which cycles, files and variables are needed?      |
| 1    | Who meets the age rule?                            |
| 2    | Who has a computable Fatty Liver Index?            |
| 3    | Who meets the metabolic dysfunction definition?    |
| 4    | How is the alcohol exclusion implemented?          |
| 5    | How are the four BMI and waist phenotypes defined? |
| 6    | How are covariates constructed?                    |
| 7-8  | How are mortality data linked, checked and saved?  |

## Target, source, analytic — and what each step costs

| Population      | Here                                                                   | Why it narrows             |
|-----------------|------------------------------------------------------------------------|----------------------------|
| <b>Target</b>   | adults with MASLD in the US                                            | who we want to speak about |
| <b>Source</b>   | NHANES 2007–2018 respondents                                           | who was sampled            |
| <b>Analytic</b> | survivors of every filter, with exposure <i>and</i> outcome measurable | who we can compute on      |

- ▶ Each narrowing step trades **generalizability** for **measurability**
- ▶ A drop is not automatically a bias. **What it is depends on its kind** — Week 2's five types, of which **only one is a missing-data problem**

## Part 1: From NHANES files to people

## A five-minute NHANES mental model

### **Cycle**

One two-year survey wave, such as 2011–2012

### **Component**

A topic-specific file, such as demographics, body measurements or laboratory tests

### **Variable**

A column inside a component, such as age or waist circumference

### **SEQN**

The participant identifier used to join files

This reproduction combines six cycles from 2007–2008 through 2017–2018.

## Stack cycles first, then join components

For each component:

1. read the same variables from all six cycles
2. stack the cycles into one long file
3. join components by SEQN

```
demo_all <- bind_rows(demo_E, demo_F, demo_G, demo_H, demo_I, demo_J)  
dat <- left_join(demo_all, bmx_all, by = "SEQN")
```

**The join choice defines which people can remain in the analysis.**

## An absent variable can look like nonresponse

Padding an unavailable variable with NA allows the code to run, but two situations then look identical:

- ▶ the question did not exist in that cycle
- ▶ the question existed and the participant has no value

Example: RIDRETH3, which identifies non-Hispanic Asian participants, begins in 2011–2012. Earlier cycles cannot supply the same classification.

**Check variable availability by cycle before interpreting an NA.**

## Part 2: Eligibility as executable logic

## The age sentence leaves a real decision

The paper states:

*"The study inclusion criteria were participants aged >18 years with evidence of MASLD."*

The reproduction uses:

```
filter(age >= 18)
```

Because NHANES age is recorded in whole years:

- ▶ age > 18 means age 19 and older
- ▶ age >= 18 includes people aged exactly 18

## The age sensitivity gives conflicting evidence

| Rule                            | Final N      | Deaths     | Mean age, SD      |
|---------------------------------|--------------|------------|-------------------|
| Published paper                 | 6,300        | 585        | 51.0, 16.6        |
| Reproduction with age $\geq 18$ | 6,371        | 586        | <b>51.0, 16.6</b> |
| Reproduction with age $> 18$    | <b>6,301</b> | <b>585</b> | 51.4, 16.3        |

Seventy reproduced participants are exactly 18 years old.

All reproduction rows are the **full reconstruction** row set (not the fasting-weight domain).

**The strict rule matches N and deaths. The inclusive rule matches the age distribution.**

## Exclusions are scientific claims

- ▶ Every filter answers “*who is this study about?*” — so every filter needs a **stated rationale**, not just a line of code
- ▶ The reproducibility contract, in one rule:

```
# Paper, Methods: eligibility stated as "aged > 18 years"  
d <- filter(d, age >= 18)      # NOT what the sentence says - deliberate
```

- ▶ A comment naming **the sentence you are implementing** lets an auditor trace every `filter()` back to the paper, and makes a **divergence visible instead of invisible**

## Same variable, three studies

Week 2 put **three** age rules on one slide:

| Study                  | Rule      | Means       |
|------------------------|-----------|-------------|
| Thursday's lab (L2)    | $\geq 20$ | 20 and over |
| This paper, as printed | $> 18$    | 19 and over |
| Our reproduction       | $\geq 18$ | 18 and over |

NHANES age is an integer, so one variable yields three different studies.

*The discipline is not to pick the right one. It is to **record which one you used**, so a reader can undo it. A reproduction that quietly adopts a different rule has destroyed its own evidence.*

## Part 3: Turning MASLD into data

## The paper constructs MASLD from measurements

In this study, a participant must have:

1. **hepatic steatosis**, defined using a Fatty Liver Index (FLI) of at least 60
2. **metabolic dysfunction**, defined by at least one of five criteria

NHANES does not provide one MASLD diagnosis column for these six cycles. The analyst constructs the case definition.

2017–2018 includes liver transient elastography, but no comparable liver-imaging measure spans all six cycles used by this paper.

## The Fatty Liver Index uses four inputs

$$FLI = 100 \frac{\exp(L)}{1 + \exp(L)}$$

$$L = 0.953 \ln(TG) + 0.139(BMI) + 0.718 \ln(GGT) + 0.053(waist) - 15.745$$

| Input                            | NHANES variable | Unit              |
|----------------------------------|-----------------|-------------------|
| Triglycerides (TG)               | LBXTR           | mg/dL             |
| Body mass index (BMI)            | BMXBMI          | kg/m <sup>2</sup> |
| Gamma-glutamyl transferase (GGT) | LBXSGTSI        | U/L               |
| Waist circumference              | BMXWAIST        | cm                |

One participant can sit on either side of the FLI threshold

Hold BMI at 28.5 kg/m<sup>2</sup>, GGT at 30 U/L and waist at 99 cm:

| Triglycerides | FLI  | Steatosis by this rule? |
|---------------|------|-------------------------|
| 150 mg/dL     | 66.4 | Yes                     |
| 110 mg/dL     | 59.5 | No                      |

The cut occurs at approximately **112.4 mg/dL** for this profile.

**A continuous score becomes a hard inclusion boundary.**

## Fasting design and item missingness are different

Triglycerides come from a fasting laboratory subsample with its own weight, WTSAF2YR.

| Stage after age eligibility       | N      | Meaning                              |
|-----------------------------------|--------|--------------------------------------|
| Positive fasting subsample weight | 14,962 | Design-valid fasting domain          |
| All four FLI inputs available     | 14,170 | 792 additional records lack an input |

Assignment to the fasting subsample is design-based. Participation, fasting compliance and laboratory nonresponse can still affect who has a positive weight and complete data.

## Two analytic row sets answer different questions

|                    | Full reconstruction               | Fasting-weight domain            |
|--------------------|-----------------------------------|----------------------------------|
| FLI computable     | 14,989                            | 14,170                           |
| Final              | <b>6,371</b>                      | <b>6,048</b>                     |
| mortality-linked N |                                   |                                  |
| Main use           | Reproduce the paper-like analysis | Design-aware population analysis |

The 323-person difference is not an arithmetic error.

**Every reported N should name its row set.**

## Metabolic dysfunction requires at least one of five criteria

| Criterion       | Operational rule                                                          |
|-----------------|---------------------------------------------------------------------------|
| Adiposity       | BMI $\geq 25$ , or waist $> 94$ cm in men or $> 80$ cm in women           |
| Glycaemia       | glucose $\geq 100$ , HbA1c $\geq 5.7\%$ , diabetes diagnosis or treatment |
| Blood pressure  | SBP $\geq 130$ , DBP $\geq 85$ or treatment                               |
| Triglycerides   | TG $\geq 150$ or lipid-lowering treatment                                 |
| HDL cholesterol | $\leq 40$ in men or $< 50$ in women                                       |

The paper also mentions treatment targeting HDL. The supplied implementation has no separate variable for that clause.

## Two participants meet none of the five criteria

Both have FLI-defined steatosis, but no operational cardiometabolic criterion:

| SEQN  | BMI, waist    | Glucose, HbA1c | BP     | TG, HDL  |
|-------|---------------|----------------|--------|----------|
| 58194 | 22.2, 87.2 cm | 94, 5.6%       | 120/67 | 108, 141 |
| 71217 | 24.9, 93.3 cm | 99, 5.0%       | 109/62 | 135, 47  |

Therefore, the MASLD step changes **6,376 to 6,374** in the full reconstruction.

An unresolved criterion contributes no positive count. Check whether each NA is a structural skip, cycle absence, refusal or genuine missingness before interpreting it.

Part 4: A Methods sentence can produce several cohorts

## The alcohol sentence is not executable as written

The paper excludes consumption above two drinks per day for men and one for women.

The sentence leaves three questions:

- ▶ Does “per day” mean per drinking day or an average over all days?
- ▶ Which variables supply frequency and quantity?
- ▶ What should happen when an alcohol response is missing?

ALQ130 records drinks **on drinking days**, not average drinks per calendar day.

## Three implementations produce three cohorts

| Implementation                               | Removed | Final N      |
|----------------------------------------------|---------|--------------|
| Threshold ALQ130 directly                    | 2,063   | 4,308        |
| Frequency $\times$ quantity averaged per day | 427     | 5,944        |
| Alcohol exclusion switched off               | 0       | <b>6,371</b> |
| Published paper                              |         | <b>6,300</b> |

Counts are the **full reconstruction** row set.

Among the tested alcohol implementations, switching the exclusion off comes closest to the published cohort. It does not prove what the authors implemented.

The age  $>18$  sensitivity, not the alcohol rule, explains most of the remaining 71-person gap.

## Reserved codes can look like extreme measurements

For ALQ130:

| Code | Meaning     |
|------|-------------|
| 777  | Refused     |
| 999  | Do not know |

```
qty <- ifelse(ALQ130 %in% c(777, 999), NA, ALQ130)
```

Leaving these codes in reclassifies nine refusals as extremely heavy drinkers.

**Missing-value codes are variable-specific. Always check the codebook.**

## The analysis switch records the decision

```
APPLY_ALCOHOL_EXCL <- FALSE
```

The pipeline still:

- ▶ constructs the alcohol variable
- ▶ saves the heavy-drinking flag
- ▶ reports the alternative cohort size
- ▶ writes the switch setting into the funnel

**A visible switch preserves the unresolved question for the next analyst.**

# BREAK

Restart question

Which alcohol implementation would you defend, and what would you write in the Methods?

## Part 5: Exposure, covariates and outcome

## The exposure has four BMI and waist phenotypes

Obesity means BMI  $\geq 30$  kg/m<sup>2</sup>, or  $\geq 25$  kg/m<sup>2</sup> for Asian participants. High central adiposity means waist-to-height ratio (WHtR)  $\geq 0.6$ .

|            | WHtR <0.6      | WHtR $\geq 0.6$ |
|------------|----------------|-----------------|
| Obesity    | Group I: 391   | Group II: 4,585 |
| No obesity | Group III: 787 | Group IV: 608   |

The exposure shares BMI and waist inputs with FLI. M1 asks what that structural dependence does to interpretation.

## Asian status is unavailable in the first two cycles

RIDRETH3 supplies a separate non-Hispanic Asian category only from 2011–2012 onward.

In 2007–2010:

- ▶ the available race variable combines “other race” and multiracial participants
- ▶ Asian participants cannot be isolated reliably
- ▶ the supplied code treats missing Asian status as “not Asian”

Seventeen early-cycle participants lie in the BMI range where Asian status could change their phenotype. **Seventeen is an upper bound, not a known error count.**

## The paper's BMI labels conflict with its Methods

**Methods:** obesity uses BMI  $\geq 30$  kg/m<sup>2</sup>, with the Asian-specific cut at 25.

**Table headers:** groups are labelled BMI  $\geq 25$  and BMI  $< 25$ .

**Published non-obesity medians:** 27.6 and 28.8 kg/m<sup>2</sup>.

Those medians cannot belong to groups defined entirely by BMI  $< 25$ . Coding the general cut at 30 reproduces all four published group medians.

**The table labels appear to be wrong; the Methods and reproduced distributions agree.**

## Cohort construction exposes under-specified covariates

The paper says that comorbidity definitions were “previously described,” without giving the full operational rules in this article.

The reproduction must choose definitions for hypertension, diabetes, dyslipidaemia, chronic kidney disease and medication use.

**Do not choose a definition because it makes Table 1 match.** Record the rule first, then report the residual difference.

This problem returns in **P2**, when Table 1 and confounding variables are constructed.

## Mortality comes from a separate linked file

The NCHS linked mortality files supply:

- ▶ mortality status
- ▶ underlying cause category
- ▶ follow-up time through 31 December 2019

The reproduction uses PERMTH\_EXM, months from the mobile examination centre visit, because BMI and waist were measured at that examination.

The paper does not state whether it used examination-based or interview-based follow-up.

Part 6: The funnel is an analytic result

## The fasting-weight funnel names the cause of every reduction

| Step | Criterion               | N            | Kind                  |
|------|-------------------------|--------------|-----------------------|
| 0    | NHANES 2007–2018        | 59,842       | Source frame          |
| 1    | Age $\geq 18$           | 36,580       | Eligibility           |
| 2    | Positive fasting weight | 14,962       | Design domain         |
| 3    | FLI computable          | 14,170       | Item missingness      |
| 4    | FLI $\geq 60$           | 6,052        | Target population     |
| 5    | At least one criterion  | 6,050        | Target population     |
| 6    | Phenotype assigned      | 6,050        | Classification check  |
| 7    | Mortality linked        | <b>6,048</b> | Outcome ascertainment |

The full reconstruction follows a different path

| Stage                  | N            |
|------------------------|--------------|
| Age $\geq 18$          | 36,580       |
| FLI computable         | 14,989       |
| FLI $\geq 60$          | 6,376        |
| At least one criterion | 6,374        |
| Phenotype assigned     | 6,374        |
| Mortality linked       | <b>6,371</b> |

This path best reproduces the paper's ordinary analyses. It fuses fasting design and data availability at the FLI step, so it should not replace the design-aware funnel.

## Why you never tune the cohort to hit the paper's N

- ▶ The temptation: quietly flip a threshold until N matches the abstract
- ▶ That is **fitting the process to the answer** — you have chosen your method using the result, and the method no longer carries evidence
- ▶ One sentence in this paper moves N by **2,063**. So an exact match is roughly as likely to mean *you guessed their ambiguity correctly* as *you implemented it correctly*

|                                      | What it demonstrates                  |
|--------------------------------------|---------------------------------------|
| Matched N, undocumented coding       | nothing — the match is unattributable |
| Mismatched N, traced to a named step | the reproduction worked               |

*A documented N you can defend beats a matched N you cannot.*

## A short decisions ledger preserves the audit trail

| Decision or limitation                   | Consequence                                        |
|------------------------------------------|----------------------------------------------------|
| Age $\geq 18$ versus $> 18$              | 70 participants; different published targets match |
| Alcohol exclusion switched off           | 427 would leave under the frequency-weighted rule  |
| Asian status unavailable in early cycles | At most 17 phenotype assignments could change      |
| Full versus fasting-weight row set       | 6,371 versus 6,048                                 |
| Examination versus interview clock       | Paper does not specify the choice                  |

**A decision without a quantified consequence is not fully documented.**

Part 7: Transfer to the rest of the course

## Thursday's L2 builds the same object from another paper

The benzodiazepine, opioid and mortality lab repeats the same mechanics:

```
dat.mort    <- rbind(mortality_cycles)
dat.nhanes  <- rbind(nhanes_cycles)
dat <- merge(dat.mort, dat.nhanes, by = "SEQN", all = TRUE)
```

Then it applies a numbered chain of eligibility rules.

**Your task in L2:** write N remaining, N removed and the reason for every step.  
*If you can only do it on MASLD, you have not learned it.*

## One more thing to notice on Thursday

L2's exposure has four levels, and the reference category is “**Neither**”:

```
dat6 <- subset(dat5, benzodiazepines == "Yes" | ssris == "Yes" | opioid  
dat6$exposure <- relevel(dat6$exposure, ref = "Neither")
```

- ▶ Eligibility required **at least one** of three drug classes. So “Neither” means *neither benzodiazepines nor opioids* — **everyone in it is an SSRI user**
- ▶ The reference group is an **active comparator**, not an unexposed population
- ▶ Exactly the shape of our Group I: every hazard ratio in the MASLD paper is measured against a group that is **obese**  
*A ratio without its referent means nothing — and in both studies the referent is not “healthy”.*

## The same cohort work returns in P1 through P5

| Increment | What this lecture contributes                               |
|-----------|-------------------------------------------------------------|
| P1        | Cohort code, decision ledger and typed funnel               |
| P2        | Operational covariates and Table 1                          |
| P3        | An interaction analysis not present in this reproduction    |
| P4        | Fasting subsample weight, strata and primary sampling units |
| P5        | The 792 missing FLI inputs and later complete-case losses   |

M1 asks which threat matters most. M2 asks what reproduced and what did not. M3 asks you to defend the decisions person by person.

## What to take into P1

For each cohort step, write four lines:

1. **Paper:** the exact statement or definition
2. **Problem:** the ambiguity or data limitation
3. **Decision:** the implemented rule and rationale
4. **Consequence:** the change in N, classification or estimate

Then label the consequence: eligibility, design, item missingness, target population or outcome ascertainment.

*Be ready to show one person who enters or leaves because of your decision.*

## Reference

Kueh MTW, Intaran MAU, Goh R, et al. **Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease: a population-based analysis of NHANES, 2007–2018.** *BMJ Open*. 2026;16:e113719.

<https://doi.org/10.1136/bmjopen-2025-113719>

Bedogni G, Bellentani S, Miglioli L, et al. **The Fatty Liver Index: a simple and accurate predictor of hepatic steatosis in the general population.** *BMC Gastroenterology*. 2006;6:33.

CDC/NCHS documentation:

- ▶ 2017–2018 liver ultrasound transient elastography
- ▶ 2017–2018 triglycerides and fasting weight
- ▶ NHANES weighting tutorial

## Appendix: component map

| Component            | Main contribution                                                    |
|----------------------|----------------------------------------------------------------------|
| DEMO                 | age, sex, race and ethnicity,<br>poverty-income ratio, survey design |
| BMX                  | BMI, waist and height                                                |
| TRIGLY and BIOPRO    | triglycerides and GGT for FLI                                        |
| GLU, GHB, HDL        | metabolic criteria                                                   |
| BPX, BPQ, DIQ        | blood pressure and treatment criteria                                |
| ALQ                  | alcohol sensitivity analysis                                         |
| NCHS mortality files | status, cause and follow-up time                                     |

## Appendix: cardiovascular outcome check

| Definition                        | Cardiovascular deaths |
|-----------------------------------|-----------------------|
| Heart plus cerebrovascular causes | 185                   |
| Heart causes only                 | <b>158</b>            |
| Published Table 2                 | <b>158</b>            |

The saved analytic variable combines heart and cerebrovascular causes. The Table 2 script creates a separate heart-only outcome, which reproduces the published event total.

**Do not reuse a derived outcome variable without checking its definition.**

## Appendix: And L2 writes a funnel — it just does not know it yet

```
dat1 <- subset(dat, RIDAGEYR >= 20) # N = 4
dat2 <- subset(dat1, RIDSTATR == "int + mec exam") # N = 4
dat3 <- subset(dat2, TAKERX_INLASTMONTH %in% c("No", "Yes")) # N = 4
dat4 <- dat3[complete.cases(dat3$BPQ020), ] # N not
dat5 <- subset(dat4, mortstat == 0 | (mortstat == 1 & permth_int > 12))
dat6 <- subset(dat5, benzodiazepines == "Yes" | ssris == "Yes" | opioid
```

- ▶ Six eligibility steps, **N after each one** — that is a funnel, written as comments
- ▶ Now type it, the way we typed ours:

|      | dat1        | dat2   | dat3        | dat4        | dat5        | dat6                 |
|------|-------------|--------|-------------|-------------|-------------|----------------------|
| kind | ELIGIBILITY | DESIGN | MISSINGNESS | MISSINGNESS | ELIGIBILITY | TARGET<br>POPULATION |

- ▶ **dat4 prints no N.** The complete-case steps are the ones that go uncounted — in this lab and in most papers

## Appendix: Reviewing someone else's eligibility

| Ask                                                | On this paper                                                                     |
|----------------------------------------------------|-----------------------------------------------------------------------------------|
| Is the case definition <b>operational</b> ?        | mostly — $FLI \geq 60$ is exact; " $\geq 1$ criterion" needs the five spelled out |
| Is every exclusion <b>executable</b> as written?   | <b>no</b> — the alcohol criterion is not                                          |
| Does the funnel's arithmetic <b>close</b> ?        | not reported; we had to rebuild it                                                |
| Is the analytic N <b>stated and reproducible</b> ? | stated; reproducible only up to the above                                         |

- ▶ Also check the tables against the Methods: here the Table 1–2 headers read "**BMI < 25**" while the Methods define obesity at  $\geq 30$ , and the non-obese groups have BMI medians of **27.6** and **28.8** — so the headers appear to be a labelling error
- ▶ **Charity and rigour together:** assume competence, and still require that the numbers reconcile

## Appendix: course resources

- ▶ Detailed recipe: `COHORT_BUILD_STEPS.md`
- ▶ Variable crosswalk: `VARIABLE_MAP.md`
- ▶ Cohort code: `02_build_analytic.R`
- ▶ Table 1 code: `03_table1.R`
- ▶ Cox model code: `04_table2_cox.R`
- ▶ Thursday lab: Accessing and preparing NHANES mortality data

The cohort counts in this deck come from the supplied reproduction pipeline. They are reproduction results, not numbers reported by the paper unless explicitly labelled.